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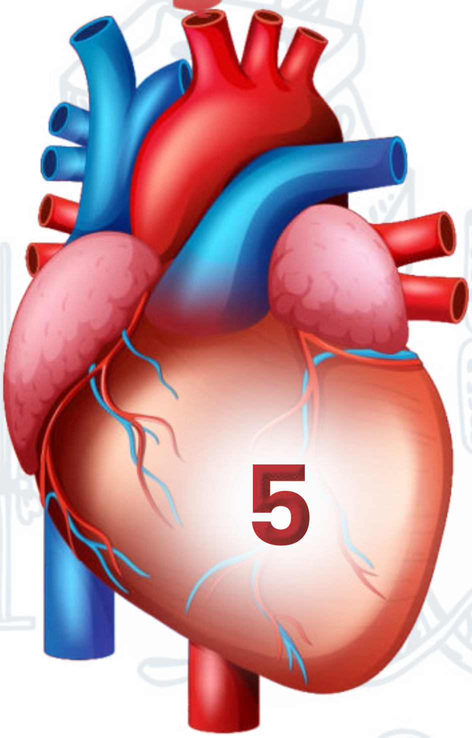

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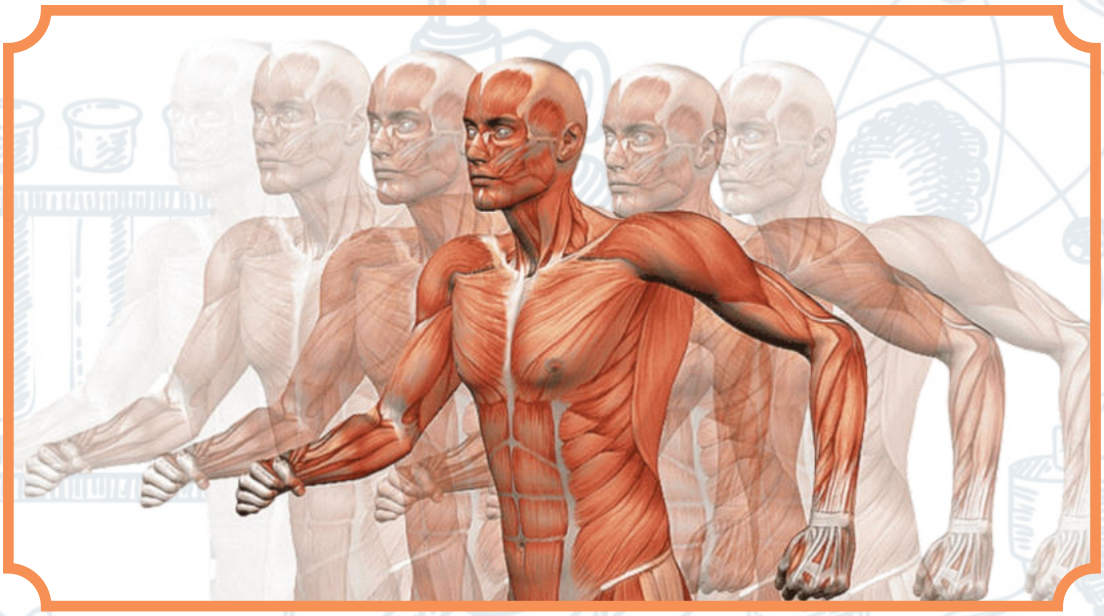
ACELLULAR LIFE



INTRODUCTION:

The cell is the basic structural and functional unit of all forms of life. Every cell consists of cytoplasm enclosed within a membrane; many cells contain organelles, each with a specific function. The term comes from the Latin word cellular meaning 'small room. Most cells are only visible under a microscope.

Viruses are small obligate intracellular parasites, which by definition contain either a RNA or DNA genome surrounded by a protective, virus-coded protein coat. Viruses may be viewed as mobile genetic elements, most probably of cellular origin and characterized by a long co-evolution of virus and host.

DISCOVERY OF A VIRUS:

•Two scientist contributed a discovery of the first virus, **"TOBACCO MOSAIC VIRUS"**

• Ivanoski reported in 1892 that extracts from infected leaves were still infectious after filtration through a chamber land filter-candle

• Bacteria are retained by such filters, a new world was discovered: filterable pathogens .
However, D.I vanovski probably did not grasp the full meaning of this discovery

•M.Beijerinck in 1898, was the first to call 'virus', the incitant of the tobacco mosaic

• He showed that the incitant was able to migrate in an ager gel, therefore being an infectious
soluble agent, or a 'contagium vivum fluidum' and definitively not a 'contagium fixum' as would be a bacteria

• Ivanovski and Beijerinck brought unequal but decisive and complementary contribution to the discovery of the viruses since then, discoveries made on Tobacco mosaic virus have stood out
as milestones of virology history.

• In 1935 W.M Stanley crystallized the infectious particle now known as Tobacco mosaic virus

•The invention of electron microscopic revealed the discovery of many viruses which being
studied under the virology

STRUCTURE OF VIRUS:

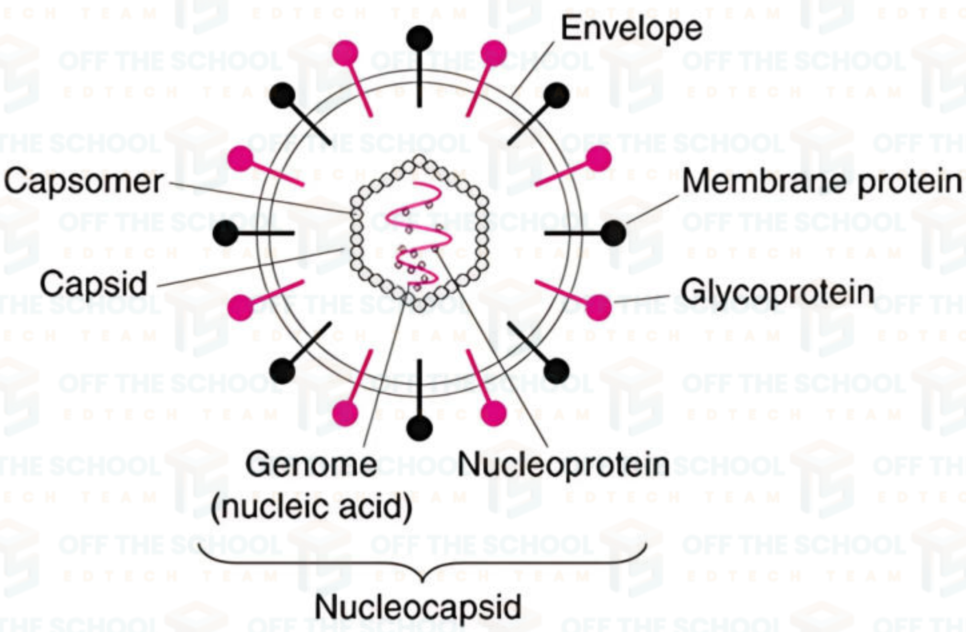
Viruses are simple, tiny infectious agents that are much smaller than bacteria, and they rely on host cells to replicate. The basic structure of a virus typically includes these components:

• **Genetic Material (Genome):** The core of a virus contains its genetic material, which can be either DNA or RNA (single-stranded or double-stranded). This genome carries the instructions for producing new viral particles and dictates the virus functions.

• **Protein Coat (Capsid):** Surrounding the genetic material is a protein coat called the capsid. The capsid protects the genome and helps the virus attach to and enter host cells. It is composed of protein subunits called capsomeres, which can arrange in various shapes (icosahedral, helical, or complex structures).

• **Envelope (Optional)**: Some viruses have an outer lipid membrane called an envelope, derived from the host cell's membrane. This envelope contains viral proteins (like glycoproteins) that help the virus bind to and enter host cells. Enveloped viruses include
influenza and HIV, while non-enveloped viruses, like adenovirus, lack this layer.

• **Surface Proteins:** These are projections on the virus's outer surface (capsid or envelope) that allow it to recognize and attach to specific receptors on the host cell surface. **Examples** include hemagglutinin and neuraminidase in influenza viruses.



Factors Affecting Virus Survival

a) Temperature

• **High Temperatures:** Most viruses are sensitive to heat. They can be destroyed quickly at higher temperatures (like above 40°C).

• **Low Temperatures:** Cold environments (like in refrigerators or freezers) can help viruses survive longer. For example, the flu virus can live for several months if kept cold.

b) Humidity (Moisture in the Air)

• **High Humidity:** Viruses with a protective outer layer (like the flu virus) may not survive well in moist conditions because their outer covering can get damaged.

• **Low Humidity:** Some viruses, like the common cold virus, survive better in dry air.

c) Sunlight (UV Light)

• Sunlight has **ultraviolet (UV) rays** that can destroy viruses by damaging their genetic material. This is why viruses often die quickly when exposed to direct sunlight.

d) Surface Type

•**Hard Surfaces (like plastic or metal):** Viruses can survive longer on smooth surfaces. For example, some viruses can live up to 2-3 days on surfaces like plastic or stainless steel.

Soft Surfaces (like fabric or paper): Viruses usually survive for shorter times on porous surfaces like cloth or paper

Classification of virus:

The Baltimore Classification system categorizes viruses based on their type of nucleic acid and their replication strategy. It was developed by Nobel Prize-winning biologist David Baltimore. Viruses are divided into seven classes:

Groups	Type of genetic material	Eukaryotic ribosomes
Prokaryotic ribosomes are smaller (70S)	eukaryotic ribosome are (80S).	eukaryotic ribosome are (80S).
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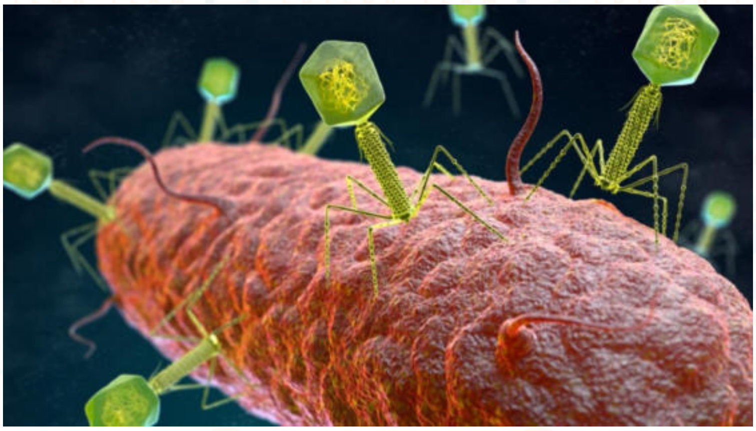
BACTERIOPHAGE VIRUS:

A bacteriophage, often simply called a phage, is a type of virus that specifically infects and kills bacteria. The term bacteriophage comes from the Greek words bakterion (meaning bacteria) and phagein (meaning to eat), so it literally means bacteria eater. These viruses are found everywhere bacteria exist, including soil, water, and the human gut.



CHARACTERISTICS OF BACTERIOPHAGE:

The nucleic acid may be either DNA or RNA and may be double-stranded or single-stranded. There are three basic structural forms of phage: an icosahedral (20-sided) head with a tail, an icosahedral head without a tail, and a filamentous form

STRUCTURE OF BACTERIOPHAGE:

Bacteriophages have a unique structure that allows them to infect bacterial cells. Here a breakdown of their main parts:

1. Head (Capsid):

o The head is a protein shell that contains the virus's genetic material.

o The genetic material can be either DNA or RNA, but most bacteriophages contain double-stranded DNA (dsDNA).

o The head is often shaped like an icosahedron (a 20-sided polygon).

2. Tail:

o The tail is a long, tube-like structure attached to the head.

o It acts like a syringe, injecting the phage's genetic material into the bacterial cell.

o The tail is often flexible, with a contractile sheath that can shorten to help the phage inject its DNA.

3. Tail Fibers:

o At the end of the tail, there are tail fibers that help the phage attach to specific receptors on the surface of the bacterial cell.

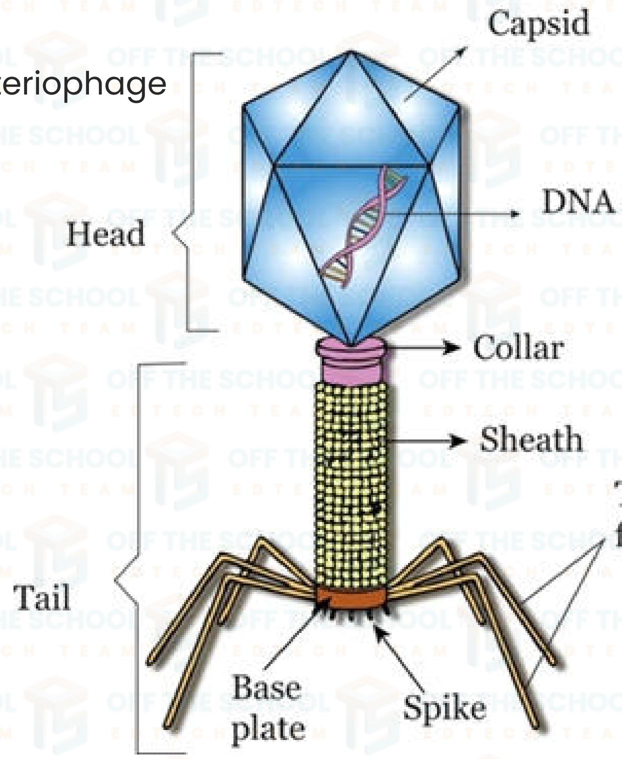
o These fibers are essential for recognizing and binding to the right type of bacteria.

LIFE CYVLE OF BACTEROIPHAGE:

There are two types of the life cycle of bacteriophage

1. Lytic cycle

2. Lysogenic cycle



Lytic cycle:

The lytic cycle is often broken into six-stages. The six stages are: attachment, penetration, transcription, biosynthesis, assembly, and lysis and release

Attachment

• The bacteriophage attaches to the surface of a bacterial cell using its tail fibers.

Penetration

• The phage injects its DNA into the bacterial cell by piercing the cell wall with its tail.

Biosynthesis

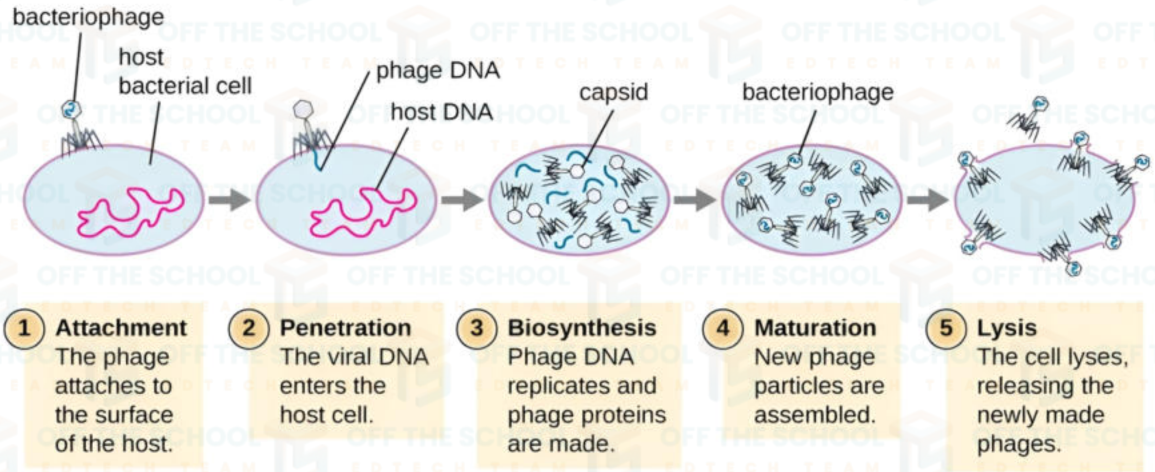
• The phage DNA takes over the bacterial machinery, forcing the cell to produce viral components (like new phage DNA and proteins).

Assembly

•New phage particles are assembled inside the bacterial cell using the newly made components.

Lysis and Release

•The bacterial cell bursts (lyses), releasing many new bacteriophages that can go on to infect other bacteria.



Lysogenic cycle:

In some cases, bacteriophages can enter a different cycle called the **lysogenic cycle:**

1. Adsorption

- The bacteriophage attaches itself to the surface of the bacterial cell.
- This attachment is highly specific, with the phage's tail fibers recognizing and binding to specific receptors on the bacterial cell wall.

2. Penetration

- After attachment, the bacteriophage injects its DNA into the bacterial cell.
- The capsid (protein coat) remains outside the bacterial cell, while the viral DNA enters the bacterial cytoplasm.

3. Integration

- Once inside the bacterial cell, the viral DNA gets incorporated into the bacterial chromosome.
- The integrated viral DNA is now called a prophage.
- This integration allows the viral DNA to be replicated along with the host's DNA during normal cell division, without causing harm to the host.

4. Replication

- As the bacterium divides, it replicates its own DNA along with the integrated prophage DNA.
- This results in the transmission of the prophage to daughter cells, allowing the virus to spread passively.
- During this period, the prophage remains dormant and does not produce new virus particles.

5. Induction

• Certain environmental factors like UV radiation, chemical agents, or stress can trigger the prophage to become active.

• The prophage DNA excises itself from the bacterial chromosome and enters the lytic cycle.

6. Synthesis

•After induction, the viral DNA takes control of the bacterial cell's machinery to produce viral components.

• This includes viral proteins and new viral DNA.

7. Assembly

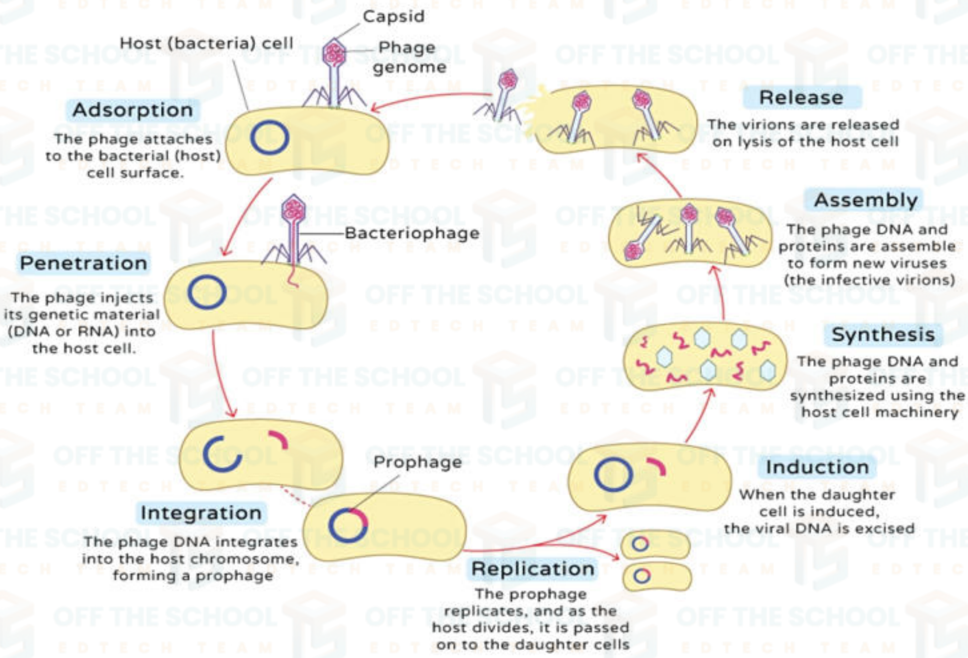
• The newly synthesized viral components are assembled into complete virions (mature virus particles).

• This includes the formation of the capsid, tail fibers, and packaging of the viral DNA into the capsid.

8. Release

•Once enough new virions are assembled, the bacteriophage produces enzymes that break down the bacterial cell wall.

• The bacterial cell bursts (lysis), releasing new bacteriophages into the environment to infect other bacterial cells.



Importance of Bacteriophages

Bacteriophages have several important uses:

1.P hage Therapy

• Used as an alternative to antibiotics to treat bacterial infections, especially antibiotic-resistant bacteria.

2. Biotechnology

• Phages are used in genetic engineering and molecular biology research.

3. Food Safety

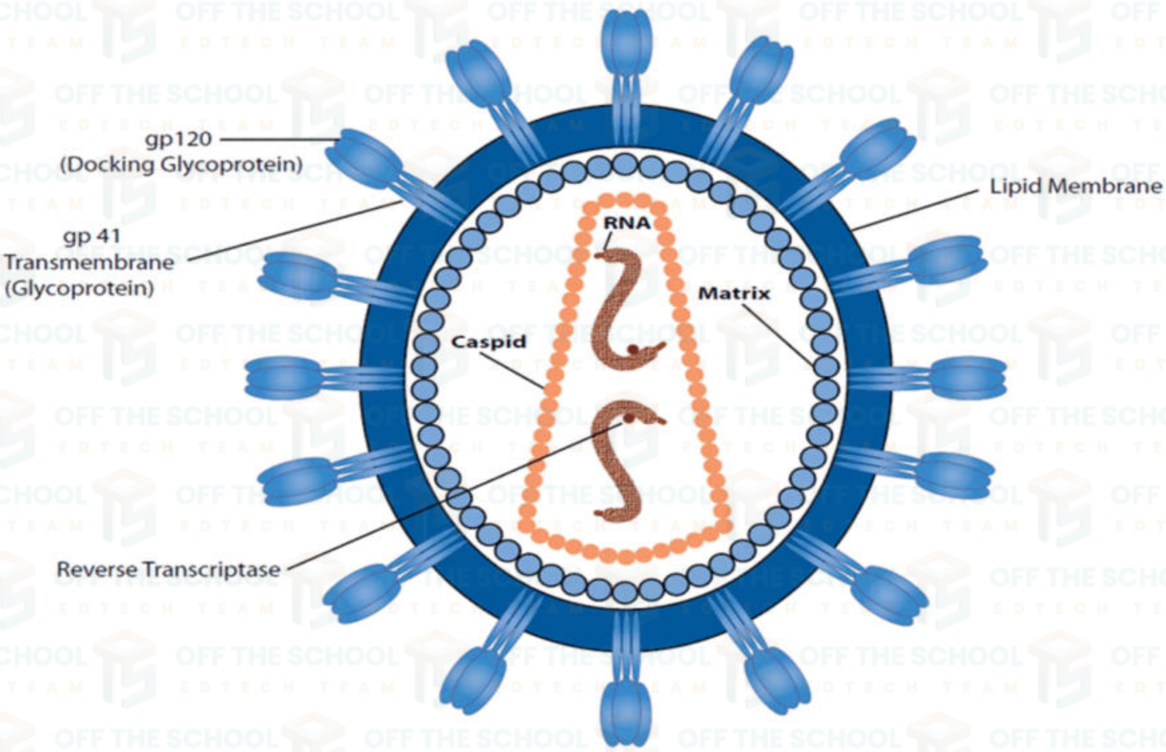
• Phages are used to kill harmful bacteria in food, helping to prevent foodborne illnesses.

4. Environmental Applications

• Phages play a role in controlling bacterial populations in natural ecosystems, like oceans and soil.

HIV VIRUS:

HIV stands for **Human Immunodeficiency Virus**. It is a virus that attacks the body immune system, specifically targeting **CD4 cells** (also known as T-helper cells), which play a crucial role in fighting infections. If left untreated, HIV can lead to **AIDS (Acquired Immunodeficiency Syndrome)**, a condition where the immune system becomes so weak that it can no longer protect the body against infections and certain cancers.



STRUCTURE OF HIV:

HIV is an **enveloped virus** with a complex structure. Here's a breakdown of its main components:

1. Envelope:

•The outer layer of the virus is a lipid membrane called the **envelope,** which comes from the host cell when the virus buds out. It is embedded with **glycoproteins** called **gp120** and **gp41** that help the virus attach to and enter host cells.

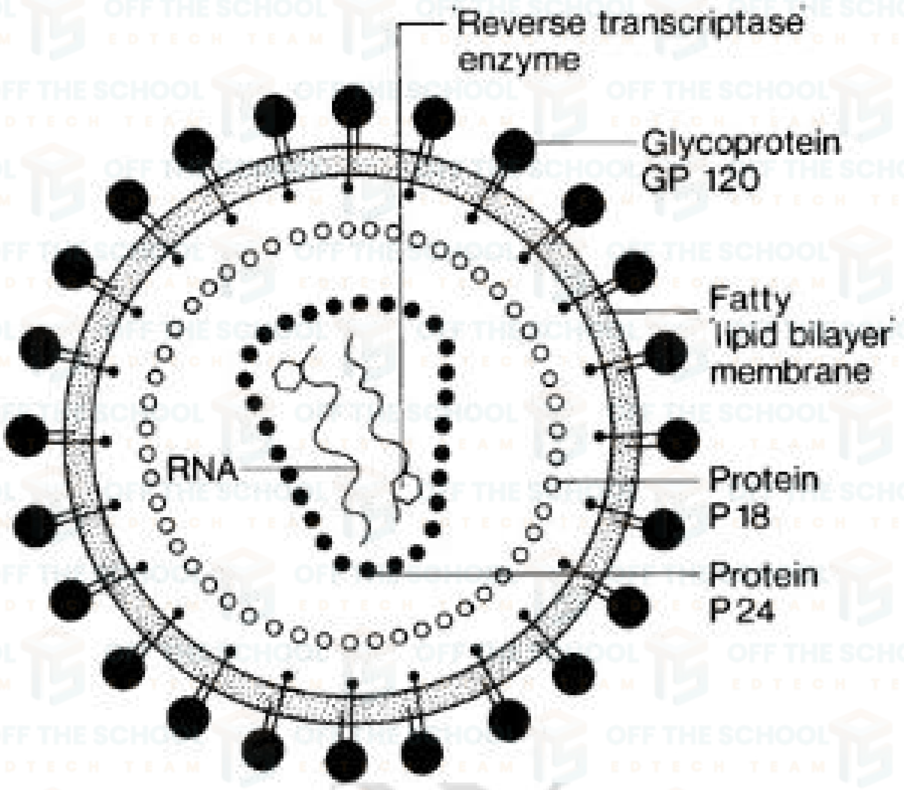
2. Capsid:

• Inside the envelope is a cone-shaped capsid made of p24 protein. This capsid encloses the viral genetic material and essential enzymes.

3. Genetic Material (RNA):

• Unlike many viruses, HIV carries its genetic information as **single-stranded RNA** (not DNA). It has two copies of RNA, making it a **diploid virus.**

• The RNA contains **9 genes** that code for 15 proteins.



4. Enzymes:

• HIV has three key enzymes necessary for its replication:

•. Reverse Transcriptase: Converts viral RNA into DNA.

• Integrase: Integrate the viral DNA in to the host's DNA

•Protease: cleaves newly synthesis viral protein to form mature virus particle

LIFE CYCLE OF HIV VIRUS:

The seven stages of the HIV life cycle are: 1) attachment, 2) fusion, 3) reverse transcription, 4) integration, 5) replication, 6) assembly, and 7) budding.

1. Attachment and Entry:

Proteins in the "tail" of the phage bind to a specific receptor on the surface of the bacterial cell

2. Fusion:

While remain attached the virus inject its RNA into the host cell

3. Reverse Transcription:

Inside the cell, HIV uses its **reverse transcriptase** enzyme to convert its RNA into **double-stranded DNA.**

4. Integration:

The newly formed viral DNA enters the cell nucleus with the help of the enzyme integrase, which **integrates** the viral DNA into the host cells DNA.

The integrated viral DNA is known as a **provirus.**

5. Replication:

The host cell machinery is tricked into reading the viral DNA, producing viral RNA and proteins needed to assemble new viruses.

6. Assembly:

New HIV particles are assembled in the host cell, including viral RNA, enzymes, and proteins.

7. Budding and Maturation:

The new viruses bud out of the host cell, taking part of the host cell membrane with them as their envelope.

Protease enzyme processes viral proteins, making the virus particles infectious.

Transmission of HIV

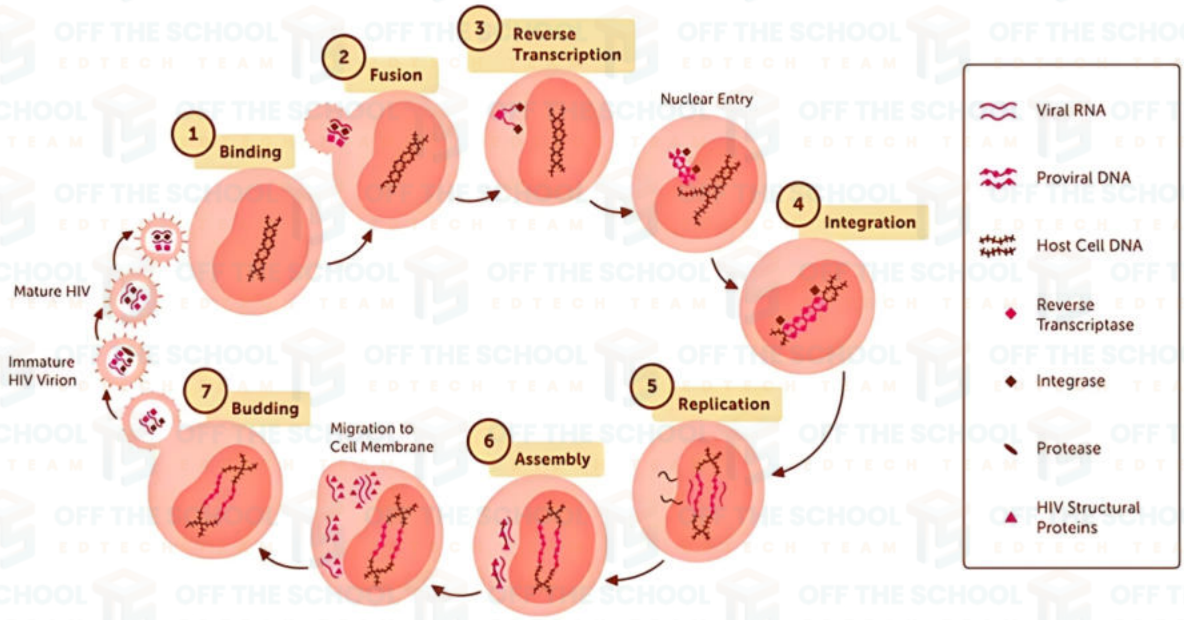
HIV can be transmitted through:

- **Sharing needles** or syringes (common among drug users)
- **Mother-to-child transmission** during pregnancy, childbirth, or breastfeeding
- **Blood transfusions** with infected blood (rare in countries with rigorous screening)
- **Exposure to infected blood** (e.g., through needle-stick injuries)

Symptoms of HIV Infection

Primary infection, also called acute HIV.

- Fever.
- Headache.
- Muscle aches and joint pain.
- Rash.
- Sore throat and painful mouth sores.
- Swollen lymph glands, also called nodes, mainly on the neck.
- Diarrhea.
- Weight loss.



TREATMENT of HIV

HIV is treated with antiretroviral medicines, which work by stopping the virus replicating in the body. This allows the immune system to repair itself and prevent further damage. A combination of HIV drugs is used Because HIV can quickly adapt and become resistant.

VIRAL DISEASES:

Viral diseases are caused by viruses. They can impact many parts of your body, including your respiratory system, digestive tract, and skin. Viral disease definition. Viruses are very small infectious agents. They are made up of a piece of genetic material, such as DNA or RNA, that enclosed in a coat of protein.

VIRAL DISEAS	CAUSITIVE VIRUS	MODE OF TRANSMISSION	MAIN SYMPTOMS	PREVENTIVE MEASURE
Influenza (Flu)	Influenza Virus (A, B, C)	Airborne droplets, direct contact	Fever, chills, muscle aches, fatigue	Annual flu vaccine, hand hygiene
COVID-19	SARS-CoV-2 (Coronavirus)	Airborne droplets, close contact	Fever, cough, loss of taste/smell, fatigue	COVID-19 vaccine, masks, social distancing
HIV/AIDS	Human Immunodeficiency Virus (HIV)	Sharing needles mother to child and sexual contents	Weight loss, chronic fatigue, infections	Get tested of HIV and do not inject drugs and others
Measles	Measles VIRUS	Airborne droplets	High fever, rash, cough, conjunctivitis	Measles vaccine (MMR vaccine)

Hepatitis B	Hepatitis B Virus (HBV)	Blood, sexual contact, mother-to-child	Jaundice, liver damage, fatigue	Hepatitis B vaccine, avoid sharing needles
Dengue Fever	Dengue Virus (DENV)	Mosquito bites (Aedes species)	High fever, severe headache, joint pain	Mosquito control, use of insect repellents
Polio	Poliovirus	Fecal-oral route, contaminated water	Paralysis, muscle weakness, fever	Polio vaccine (OPV/IPV)